

Indian Statistical Institute

Course Name: M. Math, 2nd year

Subject Name : Topology III

Exam name : HW 2

Due: August 28, 2025

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- You may use any results proved in class. Any other results (including those in homework problem sets) require proof. You are required to provide proper mathematical reasons for your answers.

1. **a)** Show that there is no 5-manifold M with $H_0(M) = \mathbb{Z}$, $H_3(M) = \mathbb{Z}/3$, $H_5(M) = \mathbb{Z}$ and $H_i(M) = 0$, $\forall i \neq 0, 3, 5$.
b) Show that there is no 3-manifold M with $H_0(M) = \mathbb{Z}$, $H_1(M) = \mathbb{Z}/2$, $H_2(M) = \mathbb{Z}/2$, and $H_i(M) = 0$, $\forall i \neq 0, 1, 2$.
2. Let M be a simply connected n -manifold. Prove that the torsion subgroups of $H_{n-1}(M)$ and $H_{n-2}(M)$ are trivial.
3. Let M be a simply connected 7-manifold. Prove that the torsion subgroup of $H_2(M)$ is isomorphic to the torsion subgroup of $H_4(M)$.